

Ch 0: Preliminaries

- Well-Ordering Principle:** Every non-empty subset of $\mathbb{Z}^+$ contains a smallest element.
- Division Algorithm:** $\forall a, b \in \mathbb{Z}, b > 0, \exists! q, r$ such that $a = bq + r, 0 \leq r < b$.
- GCD & Bézout's Identity:** $\gcd(a, b) = d \iff d = \min\{ax + by > 0 : x, y \in \mathbb{Z}\}$. a, b coprime $\iff \exists x, y : ax + by = 1$.
- Euclid's Lemma:** $p \mid ab \implies p \mid a$ or $p \mid b$ (p is prime).
- Euler's Totient $\phi(n)$:** Number of integers $1 \leq k \leq n$ with $\gcd(k, n) = 1$.

$$\phi(n) = n \prod_{p \mid n} \left(1 - \frac{1}{p}\right); \quad \sum_{d \mid n} \phi(d) = n$$

- Equivalence Relation:** Reflexive, symmetric, transitive $\implies$ Partitions the set into disjoint equivalence classes.

Ch 1 & 2: Groups & Axioms

Group Definition: A set G with binary operation $*$ satisfying:

- Closure:** $a * b \in G, \forall a, b \in G$.
- Associativity:** $(a * b) * c = a * (b * c)$.
- Identity:** $\exists! e \in G : a * e = e * a = a$.
- Inverse:** $\forall a \in G, \exists! a^{-1} : a * a^{-1} = a^{-1} * a = e$.

Standard Groups & Properties:

- $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$ under $+$.
- $U(n) = \{k \in \mathbb{Z}_n : \gcd(k, n) = 1\}$ under $\times_n$; $|U(n)| = \phi(n)$.
- $GL(n, F)$: Invertible $n \times n$ matrices over field F .
- $SL(n, F) = \{A \in GL(n, F) : \det(A) = 1\}$.
- Over $\mathbb{F}_p$: $|GL(n, \mathbb{F}_p)| = \prod_{i=0}^{n-1} (p^n - p^i)$.
- $|SL(n, \mathbb{F}_p)| = \frac{|GL(n, \mathbb{F}_p)|}{p-1}$.
- Socks-Shoes Property:** $(ab)^{-1} = b^{-1}a^{-1}$.
- If $(ab)^2 = a^2b^2 \forall a, b \in G \iff G$ is Abelian.
- If $x^2 = e \forall x \in G \implies G$ is Abelian.

Ch 3: Finite Groups & Subgroups

Order of an Element: $|a| = n \iff a^n = e$ and n is the smallest positive integer. If no such n exists, $|a| = \infty$.

- $a^k = e \implies |a| \mid k$.
- $|a| = |a^{-1}|$ and $|xax^{-1}| = |a|$ and $|ab| = |ba|$.

Subgroup Tests ($H \leq G$):

- 1-Step:** $a, b \in H \implies ab^{-1} \in H$.
- 2-Step:** $a, b \in H \implies ab \in H$ and $a \in H \implies a^{-1} \in H$.
- Finite Subgroup Test:** Non-empty finite $H \subseteq G$ is a subgroup $\iff H$ is closed under multiplication.

Important Subgroups:

- Center:** $Z(G) = \{z \in G \mid zx = xz, \forall x \in G\} \leq G$.
- Centralizer of a :** $C(a) = \{x \in G \mid xa = ax\} \leq G$.
- $Z(G) = \bigcap_{a \in G} C(a)$.

Ch 4: Cyclic Groups

$G = \langle a \rangle = \{a^n : n \in \mathbb{Z}\}$. If $|G| = n$, $G \cong \mathbb{Z}_n$; if $|G| = \infty$, $G \cong \mathbb{Z}$.

- $|a| = n \implies |a^k| = \frac{n}{\gcd(n, k)}$.
- $\langle a^i \rangle = \langle a^j \rangle \iff \gcd(i, n) = \gcd(j, n)$.
- Generators of $\mathbb{Z}_n$: k where $\gcd(k, n) = 1$. Count is $\phi(n)$.
- Fundamental Theorem of Cyclic Groups:**
 - Every subgroup of a cyclic group is cyclic.
 - If $|\langle a \rangle| = n$, for each divisor d of n , there exists a **unique** subgroup of order d , generated by $\langle a^{n/d} \rangle$.
- Elements of order d in a finite group G :**
 - If G is cyclic and $d \mid |G|$: exactly $\phi(d)$ elements.
 - If G is non-cyclic: number of elements of order d is a multiple of $\phi(d)$ (possibly 0).

Ch 5: Permutation Groups

Symmetric Group S_n ($|S_n| = n!$):

- Every $\sigma \in S_n$ is a product of disjoint cycles. Disjoint cycles **commute**.
- Order of σ :** $\text{lcm}(\text{lengths of disjoint cycles})$.
- Every permutation is a product of 2-cycles (transpositions).
- Parity:** A permutation is even/odd if written as an even/odd number of 2-cycles. Parity is invariant.

Alternating Group A_n :

- Set of even permutations; $|A_n| = n!/2$.
- r -cycle is **even** $\iff r$ is **odd**.
- A_n is generated by 3-cycles.
- A_n is simple $\forall n \geq 5$ (has no non-trivial normal subgroups).

Dihedral Group D_n : Group of symmetries of regular

n -gon. $|D_n| = 2n$.

$$D_n = \langle r, s \mid r^n = 1, s^2 = 1, srs = r^{-1} \rangle$$

- Contains n rotations $\langle r \rangle \cong \mathbb{Z}_n$ and n reflections (sr^k , all order 2).
- $Z(D_n) = \{e\}$ if n is odd; $Z(D_n) = \{e, r^{n/2}\}$ if n is even.

Ch 6: Isomorphisms

$\phi : G \rightarrow \bar{G}$ bijective and $\phi(ab) = \phi(a)\phi(b)$. **Properties Preserved under Isomorphism:** $|G| = |\bar{G}|$, Abelian/Cyclic, orders of elements, number of subgroups of order d .

Key Theorems:

- Cayley's Theorem:** Every group G is isomorphic to a subgroup of $S_{|G|}$.
- Automorphism Group $\text{Aut}(G)$:** Group of isomorphisms from $G \rightarrow G$.
- $\text{Aut}(\mathbb{Z}_n) \cong U(n)$, hence $|\text{Aut}(\mathbb{Z}_n)| = \phi(n)$.
- $\text{Aut}(\mathbb{Z}) \cong \mathbb{Z}_2$.
- Inner Automorphisms:** $\text{Inn}(G) = \{\phi_g : \phi_g(x) = gxg^{-1}\} \leq \text{Aut}(G)$.
- $\text{Inn}(G) \cong G/Z(G)$.

Ch 7: Cosets & Lagrange's Theorem

Coset Properties ($H \leq G$): $aH = \{ah : h \in H\}$.

- $aH = bH \iff a^{-1}b \in H \iff b \in aH$.
- $|aH| = |H| = |Ha|$.
- Cosets partition G .

Lagrange's Theorem: If G is finite and $H \leq G$, then $|H|$ divides $|G|$.

$$[G : H] = \frac{|G|}{|H|} \quad (\text{Index of } H \text{ in } G)$$

- Corollaries:** $|a|$ divides $|G|$; Groups of prime order p are cyclic $\implies G \cong \mathbb{Z}_p$.
- Fermat's Little Theorem:** $a^p \equiv a \pmod{p}$.
- Euler's Theorem:** $a^{\phi(n)} \equiv 1 \pmod{n}$ if $\gcd(a, n) = 1$.
- Subgroup product formula: $|HK| = \frac{|H||K|}{|H \cap K|}$. ($HK \leq G \iff HK = KH$).

Ch 8: External Direct Products

$$G_1 \oplus G_2 \oplus \dots \oplus G_n = \{(g_1, \dots, g_n) : g_i \in G_i\}.$$

- $|(g_1, g_2, \dots, g_n)| = \text{lcm}(|g_1|, |g_2|, \dots, |g_n|)$.
- $\mathbb{Z}_{mn} \cong \mathbb{Z}_m \oplus \mathbb{Z}_n \iff \text{gcd}(m, n) = 1$.
- $U(n)$ Structure Theorem:

$$U(st) \cong U(s) \oplus U(t) \quad \text{for } \text{gcd}(s, t) = 1$$

$$U(p^k) \cong \mathbb{Z}_{p^{k-1}(p-1)} \quad (p > 2 \text{ is prime})$$

$$U(2) \cong \{0\}, \quad U(4) \cong \mathbb{Z}_2$$

$$U(2^k) \cong \mathbb{Z}_2 \oplus \mathbb{Z}_{2^{k-2}} \quad (k \geq 3)$$

Ch 9: Normal Subgroups & Factor Groups

Normal Subgroup ($N \triangleleft G$): $gng^{-1} \in N, \forall g \in G, n \in N \iff gN = Ng$.

- If $[G : H] = 2 \implies H \triangleleft G$.
- If H is the **only** subgroup of order k in $G \implies H \triangleleft G$ (characteristic subgroup).
- $Z(G) \triangleleft G$.

Factor (Quotient) Groups (G/N): $(aN)(bN) = abN$. Well-defined $\iff N \triangleleft G$.

- $|G/N| = [G : N] = |G|/|N|$.
- **$G/Z(G)$ Theorem:** If $G/Z(G)$ is cyclic, then G is Abelian (hence $G/Z(G)$ is trivial).
- **Cauchy's Theorem (Abelian):** If $p \mid |G|$ (p prime), $\exists a \in G$ with $|a| = p$.

Ch 10: Homomorphisms

$\phi : G \rightarrow \bar{G}$ such that $\phi(xy) = \phi(x)\phi(y)$.

- $\ker \phi = \{x \in G \mid \phi(x) = \bar{e}\} \triangleleft G$.
- $\phi(G) \leq \bar{G}$.
- ϕ is injective $\iff \ker \phi = \{e\}$.
- $|\phi(a)|$ divides $|a|$.

Isomorphism Theorems:

- **First Isomorphism Theorem (FIT):**

$$G/\ker \phi \cong \phi(G) = \text{Im}(\phi)$$

- **Second (Diamond):** $K \leq G, N \triangleleft G \implies \frac{K}{K \cap N} \cong \frac{KN}{N}$.
- **Third:** $H, K \triangleleft G$ with $K \subseteq H \implies \frac{G/K}{H/K} \cong \frac{G}{H}$.

Homomorphism Counting Formulas:

- $|\text{Hom}(\mathbb{Z}_m, \mathbb{Z}_n)| = \text{gcd}(m, n)$.
- Trivial homomorphism is always counted.
- Homomorphisms are determined entirely by generators.

Ch 11: Fundamental Thm of Finite Abelian Groups

Classification Theorem: Every finite abelian group G is isomorphic to a direct product of cyclic groups of prime-

power order:

$$G \cong \mathbb{Z}_{p_1^{k_1}} \oplus \mathbb{Z}_{p_2^{k_2}} \oplus \dots \oplus \mathbb{Z}_{p_r^{k_r}}$$

- **Number of Non-Isomorphic Abelian Groups of Order n :** If $n = p_1^{r_1} p_2^{r_2} \dots p_k^{r_k}$, then:

$$\#(n) = p(r_1) \cdot p(r_2) \dots p(r_k)$$

where $p(r)$ is the integer partition function.

- *Partitions:* $p(1) = 1, p(2) = 2, p(3) = 3, p(4) = 5, p(5) = 7$.

High-Yield JAM Quick Reference

- **Order p :** Cyclic $\implies \mathbb{Z}_p$.
- **Order p^2 :** Always Abelian $\implies \mathbb{Z}_{p^2}$ or $\mathbb{Z}_p \oplus \mathbb{Z}_p$.
- **Order $2p$ ($p > 2$):** $\mathbb{Z}_{2p}$ or D_p .
- **Order pq ($p < q$):**
 - If $p \nmid (q-1) \implies \mathbb{Z}_{pq}$ (strictly cyclic).
 - If $p \mid (q-1) \implies \mathbb{Z}_{pq}$ or non-abelian semi-direct product.
- **Class Equation:** $|G| = |Z(G)| + \sum [G : C(x_i)]$.
- p -group has non-trivial center: $|Z(G)| > 1$ if $|G| = p^k$.